Cloud VPC

1. Log in to your AWS account, open the VPC dashboard, and create a new VPC named InstaVibeVPC with a custom IPv4 CIDR block (e.g., 10.0.0.0/16).
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   1. Log in to the **AWS Management Console.**
   2. Search for **VPC** and open the **VPC Dashboard.**
   3. Click **Create VPC**.
   4. Select **VPC only.**
   5. Enter the following details:
      - **Name:** InstaVibeVPC
      - **IPv4 CIDR Block:** 10.0.0.0/16
   6. Click **Create VPC**.
   7. Verify that the VPC has been created successfully.

2. Inside your InstaVibeVPC, create two subnets: one public subnet (e.g., 10.0.1.0/24) for hosting a web server and one private subnet (e.g., 10.0.2.0/24) for backend services.

Hint: Name your subnets as 'PublicSubnet-Zomato' and 'PrivateSubnet-Zomato' for clarity.

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Subnet Name	CIDR Block	Purpose
PublicSubnet-Zomato	10.0.1.0/24	Hosts public web servers that users can access from the internet.
PrivateSubnet-Zomato	10.0.2.0/24	Hosts backend services and databases that should not be directly accessible from the internet.

1. Open **Subnets** in the VPC Dashboard.
2. Click **Create Subnet**.
3. Select **InstaVibeVPC**.
4. Create **PublicSubnet-Zomato** with CIDR **10.0.1.0/24.**
5. Create **PrivateSubnet-Zomato** with CIDR **10.0.2.0/24**.

3. Configure an Internet Gateway and attach it to your InstaVibeVPC so that only the public subnet can access the internet, while the private subnet cannot.

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1. Open **Internet Gateways.**
2. Click **Create Internet Gateway**.
3. Name it **InstaVibe-IGW**.
4. Click **Create.**
5. Select the Internet Gateway.
6. Click **Actions → Attach to VPC.**
7. Select **InstaVibeVPC** and attach it.

➢ This Internet Gateway allows internet access for resources in the public subnet only.

4. Set up route tables so that your public subnet routes traffic to the Internet Gateway, but your private subnet does not. Verify this by checking the route table associations.

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Public Route Table

1. Create a new Route Table named **PublicRouteTable.**
2. Associate it with **PublicSubnet-Zomato**.
3. Add the following route:

Destination	**Target**
0.0.0.0/0	Internet Gateway (InstaVibe-IGW)

Private Route Table

1. Create another Route Table named **PrivateRouteTable.**
2. Associate it with **PrivateSubnet-Zomato.**
3. Keep only the default local route. Do **not** add an Internet Gateway route.

Verification

- **PublicSubnet-Zomato** can access the internet.
- **PrivateSubnet-Zomato** cannot access the internet directly.

5. Draw a simple diagram (hand-drawn or using any online tool) showing your VPC, public subnet, private subnet, and Internet Gateway, and briefly explain in 3-4 lines how this setup is similar to how apps like Zomato separate their public user interface from their private databases.

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- **Explanation**
  - This VPC design separates public-facing resources from private backend resources. The **PublicSubnet-Zomato** hosts web servers that users can access over the internet through the Internet Gateway. The **PrivateSubnet-Zomato** contains backend services or databases that are isolated from direct internet access, improving security. Applications such as **Zomato** use a similar architecture to protect sensitive customer and business data while allowing users to access the website or mobile application securely.